

Part 1 – Business Logic Validation

Test Case ID	Module	Feature	Business Rule	Scenario	Input	Expected Result
BL-001	Auth	Admin Login	Only vishnureddi04@gmail.com + viSHNU@19696 accepted	Valid admin credentials	email=vishnureddi04@gmail.com, password=viSHNU@19696, role=admin	Login success, swimmer_id=ADMIN
BL-002	Auth	Admin Login	Reject all other credentials for admin role	Invalid admin password	email=vishnureddi04@gmail.com, password=wrong, role=admin	401 error
BL-003	Auth	Admin Login	Swimmer credentials cannot access admin	Swimmer email with admin role	email=swimmer@email.com, password=any, role=admin	401 error
BL-004	Auth	Swimmer Login	Match by parent_email or parent_mobile	Valid swimmer email login	email=registered@email.com, password=correct	Login success with swimmer_id
BL-005	Auth	Swimmer Login	Normalized mobile matching	Mobile with +91 prefix	email=+91 98765 11111, password=any	Match after stripping +91 and spaces
BL-006	Auth	Swimmer Login	Account not found	Unregistered email	email=notexist@email.com, password=any	401 with account not found message
BL-007	Auth	Swimmer Login	Password hash verification	Wrong password for existing user	email=registered@email.com, password=wrong	401 incorrect password
BL-008	Auth	Swimmer Login	Legacy accounts without password_hash	User with NULL password_hash	email=legacy@email.com, password=anything	Login success (accepts any)
BL-009	Registration	Create Account	All mandatory fields required	Missing full_name	full_name=""	400 or validation error
BL-010	Registration	Create Account	DOB must be 2010-2017	Valid DOB in range	dob=2014-03-15	Age group U-12 calculated
BL-011	Registration	Create Account	DOB out of range	DOB=2008-01-01	dob=2008-01-01	age_group=None
BL-012	Registration	Create Account	Age group calculation	Each birth year mapped	dob=2016-05-01	U-10
BL-013	Registration	Create Account	Password hashing	Password provided on signup	password=myPass123	SHA-256 hash stored
BL-014	Registration	Create Account	School OR Academy required	Neither filled	school_name="", academy_id=""	Frontend blocks submission
BL-015	Registration	Create Account	Academy ID resolution	Academy name instead of ID	academy_id="SRM Swim Club"	Resolves to ACD-001
BL-016	Registration	Create Account	Unique swimmer_id generation	New registration	Valid data	SWM-2026-XXXX format ID
BL-017	Registration	Create Account	Duplicate email	Same email registered twice	parent_email=existing@email.com	Insert succeeds (no unique constraint)
BL-018	Booking	Event Registration	Max 3 events + 1 relay	Select 4 individual events	events=[ev1,ev2,ev3,ev4]	Frontend blocks at step 3
BL-019	Booking	Fee Calculation	fee = events × fee_per_event + relay_fee + GST	3 events + relay, fee=300, relay=150	3 events, relay=true	total = (900+150) × 1.18 = 1239
BL-020	Booking	Fee Calculation	GST = subtotal × gst_rate	subtotal=900, gst_rate=0.18	From tournament record	gst=162, total=1062
BL-021	Booking	Payment	Razorpay order creation	Valid amount in paise	amount=106200 (₹1062 × 100)	order_id returned
BL-022	Booking	Payment	Payment verification	Valid signature	razorpay_order_id, payment_id, signature	verified=true
BL-023	Booking	Confirmation Email	Email sent after booking	Successful booking	Valid swimmer with parent_email	Email sent to parent_email
BL-024	Tournament	Status Update	Status changes allowed	Admin changes Open→Ongoing	status=Ongoing, tid=GNMC-2026	Status updated
BL-025	Tournament	Edit	All fields editable	Admin edits venue	venue="New Venue"	Venue updated in DB
BL-026	Tournament	Deadline Logic	Events with past deadline show "Registration	reg_deadline=yesterday	Past date	"Registration Closed" badge shown

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Closed"						
BL-027	Tournament	Wizard Filter	Only open + active deadline in wizard	3 tournaments: 1 Open active, 1 Open expired, 1 Closed	Fetch	Only 1 shown in wizard
BL-028	Heat Sheet	Generation	Swimmers sorted by seed time	10 swimmers with seed times	Generate for tournament	Fastest in centre lane
BL-029	Heat Sheet	Lane Pattern	8-lane pattern: 4,5,3,6,2,7,1,8	8 swimmers	lanes=8	Fastest gets lane 4
BL-030	Heat Sheet	Multiple Heats	More swimmers than lanes	16 swimmers, 8 lanes	Generate	2 heats created
BL-031	Results	Update Entry	Admin enters final time	entry_id, final_time=00:45.32	POST /api/results/update	final_time stored
BL-032	Certificate	Visibility	Only shown when tournament=Completed AND swimmer has booking	Completed tournament, swimmer has booking	Check profile page	Certificate card visible
BL-033	Profile	Load Data	Profile loads by swimmer_id from localStorage	swimmer_id=SWM-2026-XXXX	Navigate to player.html	All fields populated from DB
BL-034	Profile	Edit	Update swimmer fields	Change address	Address="New Address"	Address updated in DB
BL-035	Profile	Photo	Per-user photo storage	Upload photo	File < 2MB	Stored in localStorage with swimmer_id key
BL-036	Stats	Dashboard	Counts from DB	Admin views dashboard	GET /api/stats	Correct swimmer/booking/entry/academy counts
BL-037	Home	Data Caching	/api/home cached for 60 seconds	Two requests within 60s	GET /api/home twice	Second returns cached data